

Ques

Draw the graph of $y = g(x)$, where

$$g(x) = \max. \{ f(t) ; x+1 \leq t \leq x+2 \}, \quad x < 0$$
$$= 1-x, \quad x \geq 0,$$

where $f(x) = -x^2 - 2x$

Sol

$$g(x) = f(x+2), \quad x < -3$$

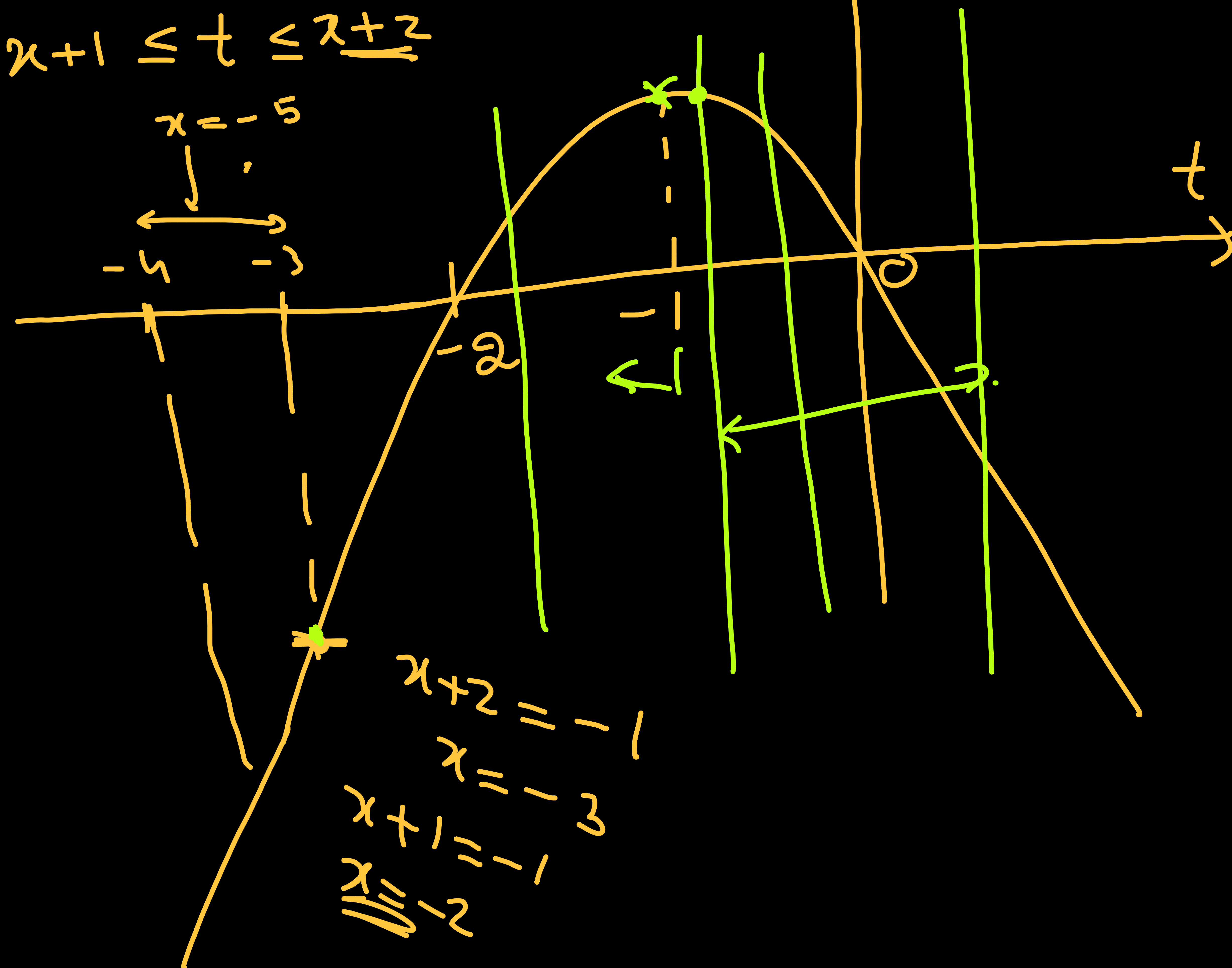
$$= f(-1), \quad -3 \leq x \leq -2.$$

$$= f(x+1), \quad -2 < x < 0$$

$$= 1-x, \quad x \geq 0.$$

$$x+1 \leq t \leq x+2$$

$$x = -5$$



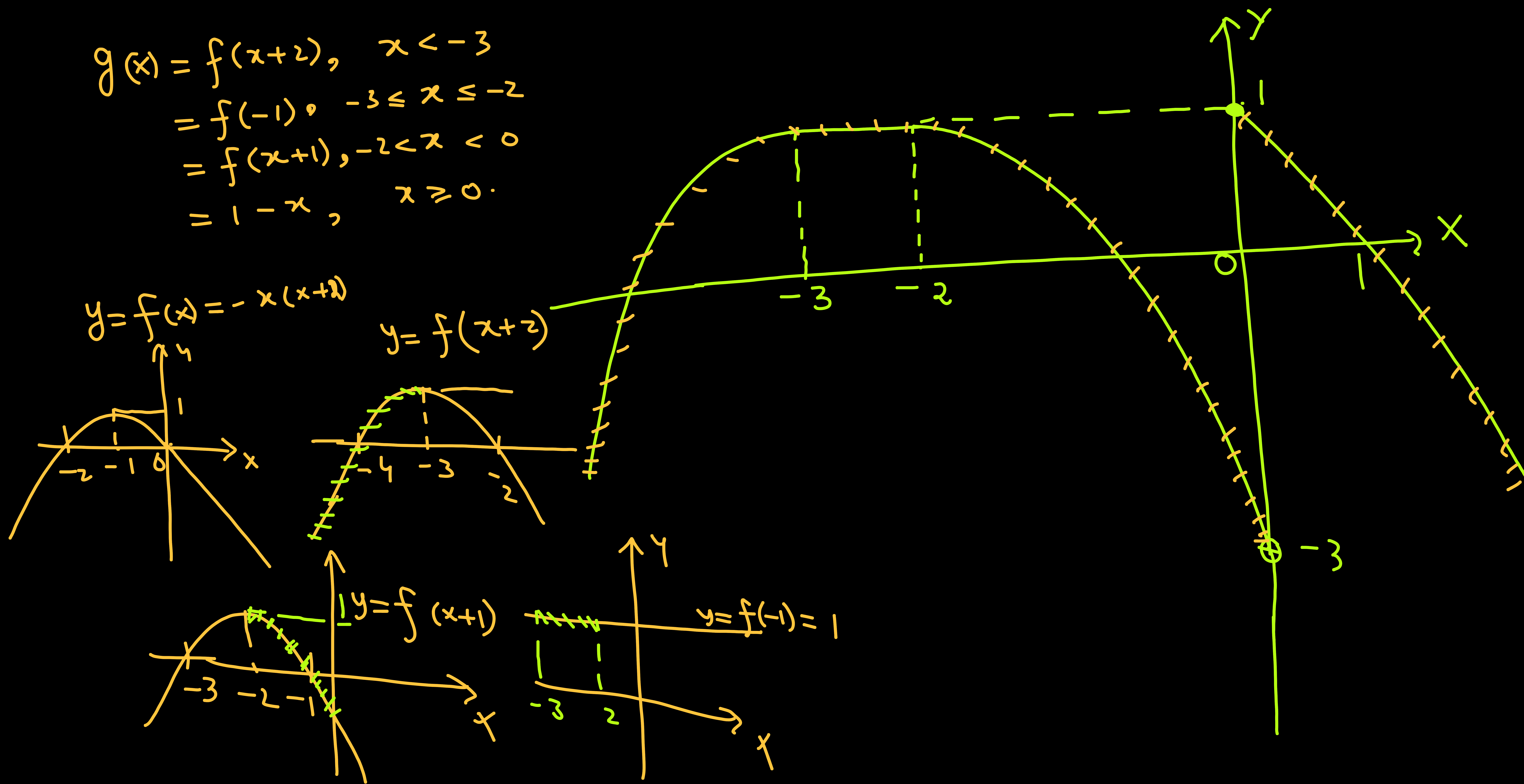
$$\begin{aligned}
 g(x) &= f(x+2), \quad x < -3 \\
 &= f(-1), \quad -3 \leq x \leq -2 \\
 &= f(x+1), \quad -2 < x < 0 \\
 &= 1-x, \quad x \geq 0.
 \end{aligned}$$

$$y = f(x) = -x(x+2)$$

$$y = f(x+2)$$

$$y = f(x+1)$$

$$y = f(-1) = 1$$



Ques Find a polynomial function $f(x)$, satisfying

$$f(2x) = f'(x) \cdot f''(x).$$

Sol

$$f(2x) = f'(x) \cdot f''(x) \Rightarrow 8ax^3 + 4bx^2 + 2cx + d = (3ax^2 + 2bx + c) \cdot (6ax + 2b)$$

$$8a = 18a^2 \Rightarrow a = \frac{4}{9}$$

$$4b = 12ab + 6ab = 18ab \Rightarrow 4b = 18 \cdot \frac{4}{9} \cdot b$$
$$\boxed{b=0}$$

$$2c = 6ac + 4b^2 \xrightarrow{b=0, a=\frac{4}{9}} \boxed{c=0}$$

$$d = 2bc \Rightarrow d = 0$$

$$\Rightarrow \boxed{f(x) = \frac{4}{9}x^3}$$

$$f(x) = ax^3 + bx^2 + cx + d$$

$$\begin{array}{ccc} f(2x) & = & f'(x) \cdot f''(x) \\ \downarrow & & \downarrow \\ n & = & n-1 + n-2 \\ & & n=3. \end{array}$$

Que If $\sqrt{x+y} + \sqrt{y-x} = 5$ then find $\frac{d^2y}{dx^2}$

Sol

$$\sqrt{x+y} + \sqrt{y-x} = 5$$

$$\Rightarrow \sqrt{x+y} = 5 - \sqrt{y-x}$$

$$x+y = 25 + y - x - 10\sqrt{y-x}$$

$$2x - 25 = -10\sqrt{y-x}$$

$$4x^2 + 625 - 100x = 100(y-x)$$

$$\therefore 4x^2 + 625 = 100y$$

$$\Rightarrow 8 = 100y'' \Rightarrow y'' = \frac{8}{100} = \boxed{\frac{2}{25}}$$

Ques Let $y = \frac{x^2}{2} + \frac{1}{2}x\sqrt{x^2+1} + \ln(x + \sqrt{x^2+1})$ satisfies
 $\lambda y = xy' + \ln(y')$, then find λ .

Sol

$$y = \frac{x^2}{2} + \frac{1}{2}x\sqrt{x^2+1} + \frac{1}{2}\ln(x + \sqrt{x^2+1})$$
$$\rightarrow y' = \lambda + \frac{\sqrt{x^2+1}}{2} + \frac{1}{2}\frac{x^2}{\sqrt{x^2+1}} + \frac{1}{2}\frac{1}{x + \sqrt{x^2+1}} \left(1 + \frac{x}{\sqrt{x^2+1}}\right)$$
$$= \lambda + \frac{1}{2}\left(\sqrt{x^2+1} + \frac{x^2}{\sqrt{x^2+1}} + \frac{1}{\sqrt{x^2+1}}\right) = \lambda + \frac{1}{2}\left(\sqrt{x^2+1} + \sqrt{x^2+1}\right) = \lambda + \sqrt{x^2+1}$$

Consider $xy' + \ln(y') = \lambda \cdot (x + \sqrt{x^2+1}) + \ln(x + \sqrt{x^2+1}) = x^2 + x\sqrt{x^2+1} + \ln(x + \sqrt{x^2+1})$

$$= 2y \Rightarrow \boxed{\lambda = 2} \quad \Delta$$

Ques Let $f(x)$ is a continuous function defined on $[-1, 1]$ and satisfies $f(2x^2-1) = 2x \cdot f(x)$, then find $f(x)$.

Soln

$$f(2x^2-1) = 2x \cdot f(x).$$

$$\downarrow x = \cos(\theta), \theta \in [0, \pi]$$

$$f(2\cos^2(\theta)-1) = 2\cos(\theta) \cdot f(\cos(\theta))$$

$$\Rightarrow f(\cos(2\theta)) = 2\cos(\theta) \cdot f(\cos(\theta))$$

$$= 2\cos(\theta) \cdot 2 \cdot \cos\left(\frac{\theta}{2}\right) \cdot f\left(\cos\left(\frac{\theta}{2}\right)\right) = 2^2 \cdot \cos(\theta) \cdot \cos\left(\frac{\theta}{2}\right) \cdot f\left(\cos\left(\frac{\theta}{2}\right)\right)$$

$$= 2^3 \cdot \cos(\theta) \cdot \cos\left(\frac{\theta}{2}\right) \cdot \cos\left(\frac{\theta}{4}\right) \cdot f\left(\cos\left(\frac{\theta}{4}\right)\right)$$

$$\vdots$$

$$= 2^{n+1} \cdot \cos(\theta) \cdot \cos\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{4}\right) \dots \cos\left(\frac{\theta}{2^n}\right) \cdot f\left(\cos\left(\frac{\theta}{2^n}\right)\right)$$

$$f(\cos(2\theta)) = 2^{n+1} \cdot \underbrace{\cos(\theta) \cos\left(\frac{\theta}{2}\right) \dots \cos\left(\frac{\theta}{2^n}\right)}_{\frac{\sin(2\theta)}{2 \sin\left(\frac{\theta}{2^n}\right)}} \cdot f\left(\cos\left(\frac{\theta}{2^n}\right)\right)$$

$$= \frac{2^{n+1}}{2^{n+1}} \cdot \frac{\sin(2\theta)}{\sin\left(\frac{\theta}{2^n}\right)} \cdot f\left(\cos\left(\frac{\theta}{2^n}\right)\right)$$

$$f(\cos(2\theta)) = \frac{\sin(2\theta)}{\sin(\frac{\theta}{2^n})} \cdot f(\cos(\frac{\theta}{2^n}))$$

$$\lim_{t \rightarrow 1} f(t) = \underline{\underline{f(1)}}$$

$$\lim_{n \rightarrow \infty} f(\cos(2\theta)) = \sin(2\theta) \cdot \lim_{n \rightarrow \infty} \frac{f(\cos(\frac{\theta}{2^n}))}{\sin(\frac{\theta}{2^n})} = \sin(2\theta) \cdot \lim_{n \rightarrow \infty} \frac{f(1 - 2\sin^2(\frac{\theta}{2^{n+1}}))}{2 \cdot \sin(\frac{\theta}{2^{n+1}}) \cdot \cos(\frac{\theta}{2^{n+1}})}$$

$$f(2x^2 - 1) = 2x \cdot f(x)$$

$$(x = -1) \Rightarrow f(2x^2 - 1) = -2x f(-x)$$

$$0 = 2x(f(x) + f(-x))$$

$\Rightarrow f(x)$ is an odd function — ①

$$x = 1 \Rightarrow f(1) = 2f(1) \Rightarrow f(1) = 0 \quad \text{②}$$

$$f(2x^2 - 1) = 2x \cdot f(x)$$

$$\Rightarrow f(x) = \frac{f(2x^2 - 1)}{2x}$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{f(2x^2 - 1)}{2x}$$

$f(0) = 0$ (As $f(x)$ is an odd continuous function)

$$f(\cos(2\theta)) = \sin(2\theta) \cdot \lim_{n \rightarrow \infty} \frac{f(1 - 2\sin^2(\frac{\theta}{2^{n+1}}))}{2\sin(\frac{\theta}{2^{n+1}}) \cdot \cos(\frac{\theta}{2^{n+1}})}.$$

$$= \sin(2\theta) \lim_{n \rightarrow \infty} \frac{f(2\sin^2(\frac{\theta}{2^{n+1}}) - 1)}{2\sin(\frac{\theta}{2^{n+1}})}$$

$$= \sin(2\theta) \cdot (0)$$

$$f(\cos(2\theta)) = 0.$$

$$\Rightarrow \boxed{f(x) = 0}, \forall x \in [-1, 1].$$

$$\lim_{t \rightarrow 0} \frac{f(2t^2 - 1)}{2t} = 0$$

Ques Find $f(x)$ satisfying
 $f(x+2y) = f(x) \cdot e^{2y} + f(2y) \cdot e^x + x^2(1-e^{2y}) + 4xy, \forall x, y \in \mathbb{R}$
 and $f'(0) = 1$ then find $f(x)$.

Sol $\xrightarrow{2y=y}$ $f(x+y) = \underline{f(x) \cdot e^y} + \underline{f(y) \cdot e^x} + \underline{x^2} - \underline{x^2 \cdot e^y} + \underline{y^2} - \underline{y^2 \cdot e^x} + 2xy$

$$f(x+y) = e^y(f(x) - x^2) + e^x(f(y) - y^2) + (x+y)^2$$

$$\Rightarrow f(x+y) - (x+y)^2 = e^y(f(x) - x^2) + e^x(f(y) - y^2)$$

$$e^{-(x+y)} [f(x+y) - (x+y)^2] = \underbrace{e^{-x}(f(x) - x^2)}_{= g(x) \text{ say}} + e^{-y}(f(y) - y^2)$$

$$\Rightarrow \begin{cases} g(x+y) = g(x) + g(y) \\ \Rightarrow g(x) = kx \end{cases}$$

Result for diff fn.

$$f(x+y) = f(x) + f(y) \\ \Rightarrow f(x) = kx$$

$$f(x+y) = f(x) \cdot f(y) \\ f(x) = e^{kx} @ 0$$

$$f(xy) = f(x) + f(y), x, y \in \mathbb{R}^+ \\ f(x) = k \ln(x)$$

$$f(xy) = f(x) \cdot f(y), \forall x, y \in \mathbb{R} \\ f(x) = x^k @ 0$$

$$g(x) = kx$$

$$e^{-x}(f(x) - x^2) = kx$$

$$f(x) = x^2 + kx \cdot e^x$$

given $f'(0) = 1$
 $\Downarrow$

$$f(x) = x^2 + x \cdot e^x$$

$$f'(x) = 2x + kx e^x + k e^x$$

$\int x=0$
 $k=1$

Result

$f(x)$ is polynomial
 & satisfies

$$f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right).$$

$\Downarrow$
 $f(x) = 1 \pm x^n$

Ques

Let $-1 \leq p \leq 1$, Then find number of real roots of the equation $4x^3 - 3x - p = 0$ in the interval $[\frac{1}{2}, 1]$.

Sol

$$x \in [\frac{1}{2}, 1]$$

$$4x^3 - 3x = p$$

$$x = \cos \theta, \theta \in [0, \frac{\pi}{3}]$$

$$4\cos^3(\theta) - 3\cos(\theta) = p$$

$$\cos(3\theta) = p$$

$$3\theta \in [0, \pi]$$

$\Rightarrow$ as $\cos(3\theta)$ can take all values from $[-1, 1]$ and in $[0, \pi]$ it will be one-one function hence

$$p \in [-1, 1]$$

for any value of 'p'

$\exists$ exactly one θ

$\Downarrow$
 $\exists$ exactly one 'x' in $[\frac{1}{2}, 1]$

$\Rightarrow$ No. of solution = $\boxed{1}$.

Alt.

$$4x^3 - 3x = p, \quad p \in [-1, 1]$$

y'' y''

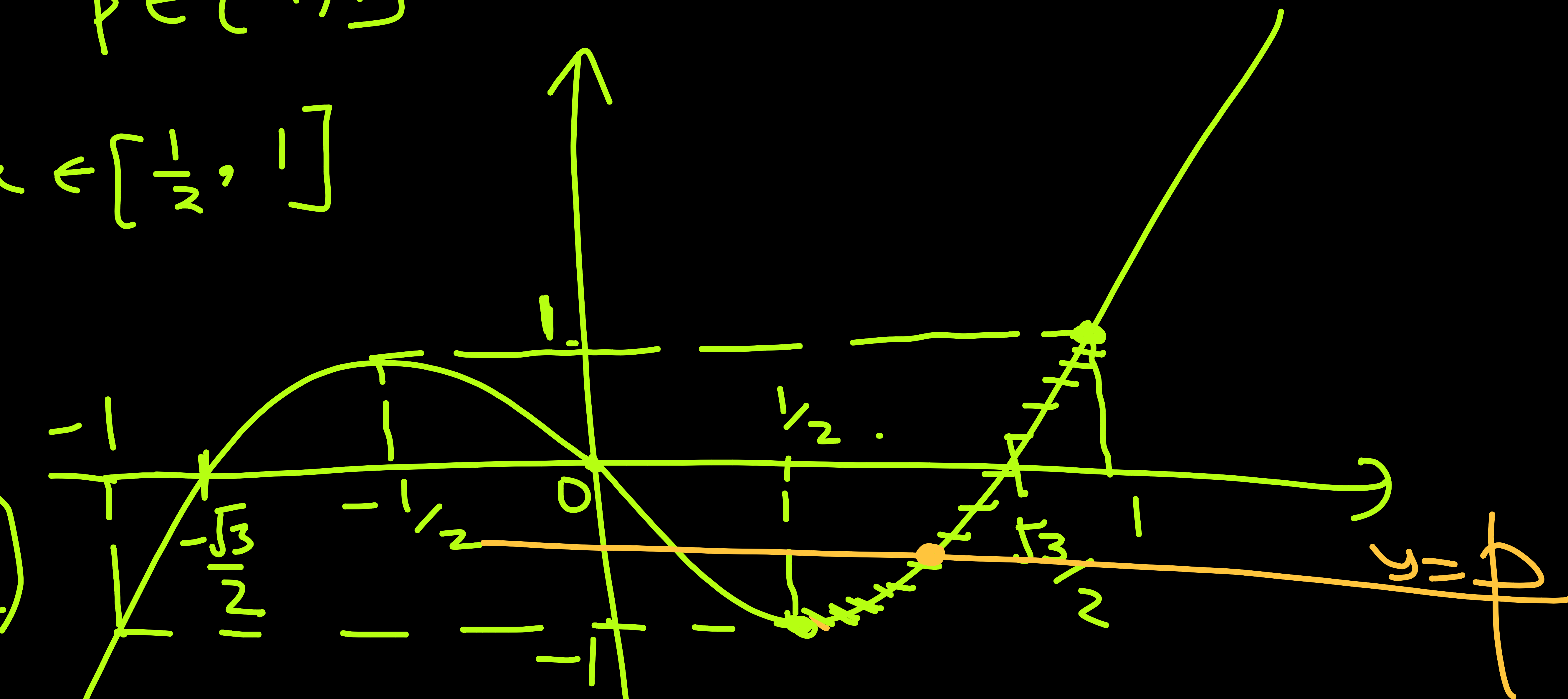
$$x \in \left[\frac{1}{2}, 1\right]$$

$$\begin{aligned} y &= 4x^3 - 3x \\ &= 4x \left(x^2 - \frac{3}{4}\right) \\ &= 4 \cdot x \left(x - \frac{\sqrt{3}}{2}\right) \left(x + \frac{\sqrt{3}}{2}\right) \end{aligned}$$

$$y' = 12x^2 - 3 = 3 \left(x^2 - \frac{1}{4}\right)$$

$$y\left(\frac{1}{2}\right) = \frac{4}{8} - \frac{3}{2} = \frac{1}{2} - \frac{3}{2} = -1$$

$$y(1) = 4 - 3 = 1$$



⇒ Required number of solutions
= 1 A

Que Let $f(x) = e^{x^2} + e^{-x^2}$, $g(x) = x \cdot e^{x^2} + e^{-x^2}$ and $h(x) = x^2 \cdot e^{x^2} + e^{-x^2}$ are 3-functions defined on $[0, 1]$ then find $(f(x))_{\max.} + (g(x))_{\max.} + (h(x))_{\max.}$

Sol

$$f(x) = e^{x^2} + e^{-x^2}, \quad x \in [0, 1]$$

$$x \in [0, 1] \Rightarrow x^2 \in [0, 1]$$

$$f(t) = e^t + e^{-t}$$

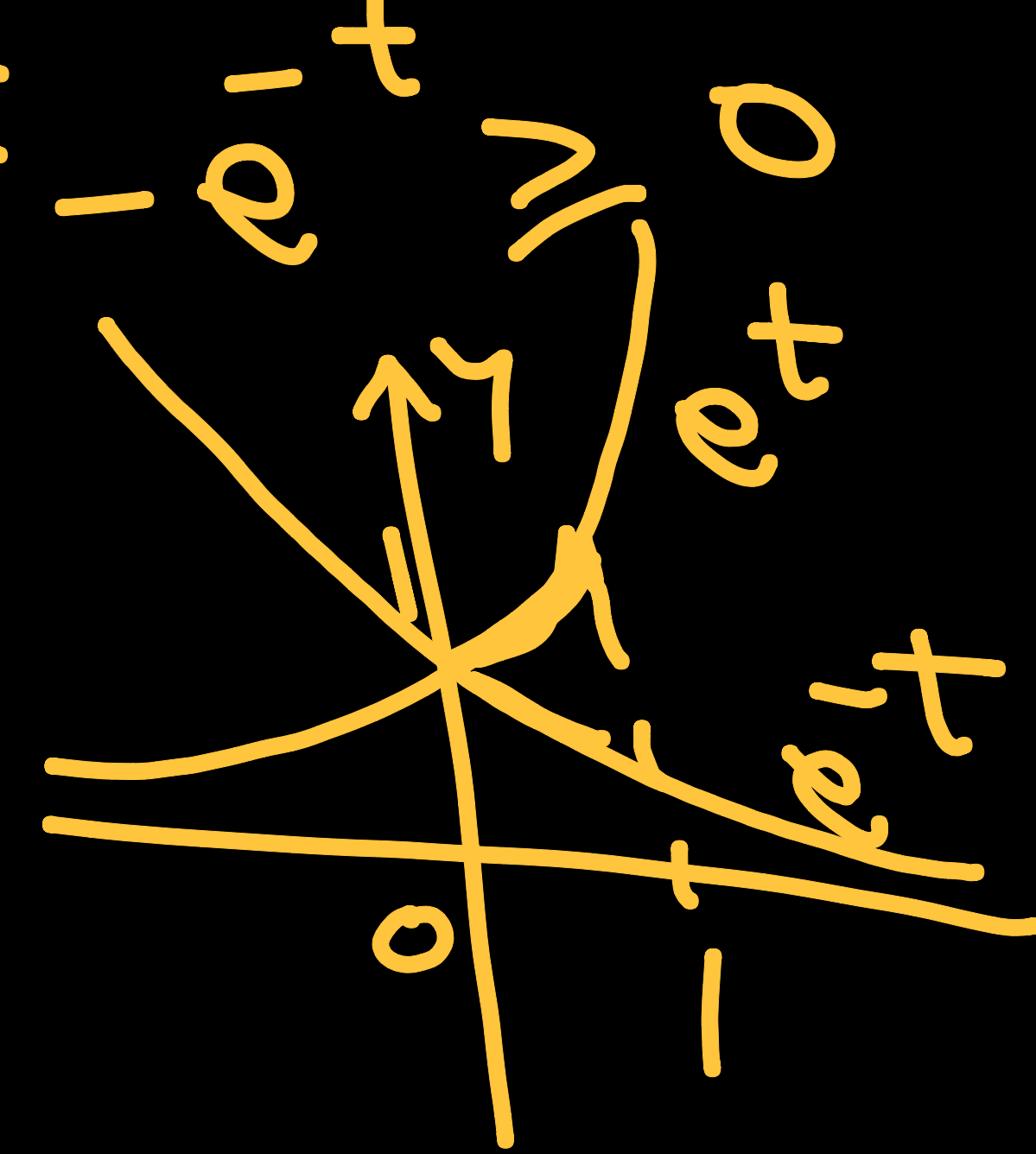
$$f'(t) = e^t - e^{-t} \geq 0$$

$\downarrow$

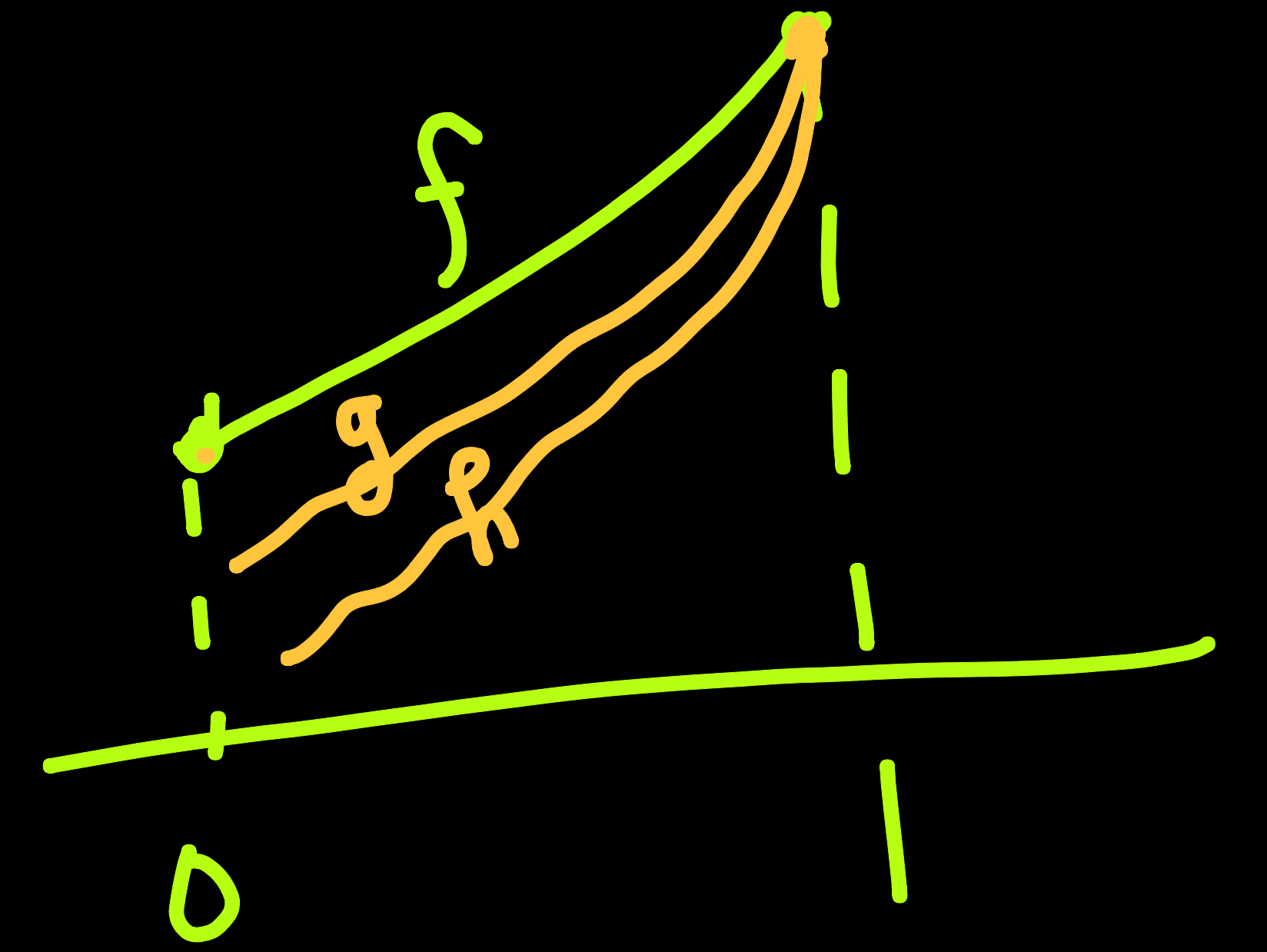
$$f(t) \uparrow$$

$$\Rightarrow f(x) \uparrow$$

$$(f(x))_{\max.} = f(1) = e + \frac{1}{e}$$



$$\left. \begin{aligned} f(x) &= e^{x^2} + e^{-x^2} \\ g(x) &= x \cdot e^{x^2} + e^{-x^2} \\ h(x) &= x^2 \cdot e^{x^2} + e^{-x^2} \end{aligned} \right\} \Rightarrow f(x) \geq g(x) \geq h(x)$$



$$(g(x))_{\max.} = g(1) = f(1)$$

$$(h(x))_{\max.} = h(1) = f(1)$$

$$(f(x))_{\max.} + (g(x))_{\max.} + (h(x))_{\max.} = 3f(1) = 3\left(e + \frac{1}{e}\right) = \frac{3(e^2 + 1)}{e}$$

Ans

Que-9 Let $g(x) = \ln(f(x))$, where $f(x)$ is a twice differentiable positive function on $(0, \infty)$ such that $f(x+1) = x \cdot f(x)$, then find value of $g''(n+\frac{1}{2}) - g''(\frac{1}{2})$ for some $n \in \mathbb{N}$.

Sol $g(x) = \ln(f(x))$

$$\begin{aligned} f(x+1) &= x \cdot f(x) \\ \Rightarrow \ln(f(x+1)) &= \ln(x) + \ln(f(x)) \\ g(x+1) &= \ln(x) + g(x) \end{aligned}$$

$$\begin{aligned} g(x+1) - g(x) &= \ln(x) \\ \Rightarrow g''(x+1) - g''(x) &= -\frac{1}{x^2} \end{aligned}$$

$$x = \frac{1}{2} \rightarrow g''(\frac{3}{2}) - g''(\frac{1}{2}) = -4$$

$$x = \frac{3}{2} \rightarrow g''(\frac{5}{2}) - g''(\frac{3}{2}) = -\frac{4}{9}$$

$$x = \frac{5}{2} \rightarrow g''(\frac{7}{2}) - g''(\frac{5}{2}) = -\frac{4}{25}$$

$$\vdots$$

$$g''(n+\frac{1}{2}) - g''(n-\frac{1}{2}) = -\frac{4}{(2n-1)^2}$$

$$+ g''(n+\frac{1}{2}) - g''(\frac{1}{2}) = -4 \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots + \frac{1}{(2n-1)^2} \right)$$

R. Am.

Ques Let $f(x) = \frac{\ln(\pi+x)}{\ln(e+x)}$, $\forall x \in \mathbb{R}^+$ then find the interval in which $f(x)$ increases.

Sol

$$f(x) = \frac{\ln(\pi+x)}{\ln(e+x)}$$

$$f'(x) = \frac{\frac{\ln(e+x)}{\pi+x} - \frac{\ln(\pi+x)}{e+x}}{(\ln(e+x))^2}$$

$$= \frac{(e+x)\ln(e+x) - (\pi+x)\ln(\pi+x)}{(\pi+x)(e+x) \cdot (\ln^2(e+x))}$$

$< 0 \Rightarrow f(x) \downarrow$

$\Rightarrow$ If $x > 0$ then

$e+x > e$ and $\pi+x > \pi$

$$(\pi+x) > (e+x)$$

$$\ln(\pi+x) > \ln(e+x) > 1$$

$$(\pi+x) \cdot \ln(\pi+x) > (e+x) \cdot \ln(e+x)$$

$\therefore$ In $\mathbb{R}^+$, $f(x)$ never increases